ABSTRACT

Student Stress Level prediction is a machine learning application designed to classify the stress level of a student as Low, Medium, or High based on daily lifestyle and academic factors such as sleep hours, study hours, screen time, physical activity, and the number of pending assignments. The main objective of this project is to develop a model that analyses these factors and predicts the likely stress category. Such a system shows how machine learning can identify relationships between student habits and stress, helping users understand the basic process of data-driven prediction.

The project is implemented using Python and its data-processing and machine-learning libraries. A dataset of 60 student records with the above features and their stress levels is prepared and processed with Pandas and NumPy. The data is divided into training and testing sets in an 80:20 ratio, and the features are standardised using StandardScaler. The training data is used to train a Logistic Regression classifier from the Scikit-learn library, which learns the relationship between the input features and the stress category. A Random Forest classifier is also trained for comparison and to identify which features influence stress the most. After training, the model is evaluated on the testing data and can predict the stress level of a new student when feature values are provided.

To examine performance, the project uses accuracy, the confusion matrix, and the classification report (precision, recall, and F1-score). Visualisations including a correlation heatmap, a confusion matrix, and a feature-importance chart support the analysis. On the 12-record test set, both Logistic Regression and Random Forest reached an accuracy of about 67%, with the Medium class being the hardest to predict. Because the dataset is small and synthetic, this result is only a demonstration of the workflow and does not show that the model will be accurate for real students.

Overall, this project demonstrates the fundamental workflow of a classification-based machine learning system, including dataset preparation, preprocessing, model training, prediction, and performance evaluation. The model can be improved in future work by using a larger real-world survey dataset, adding features such as exam pressure, mental well-being scores, and social factors, and validating it on unseen data. This would help assess whether the system can provide reliable stress-level estimates for different students. The system is intended only as an awareness tool and not as a medical or psychological diagnosis.

Keywords: Machine Learning, Student Stress Prediction, Logistic Regression, Random Forest, Python, Classification.